

Uncertainty II

STAT 240 - Fall 2025

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Interval Estimation

| All models are wrong, but some are useful - George Box

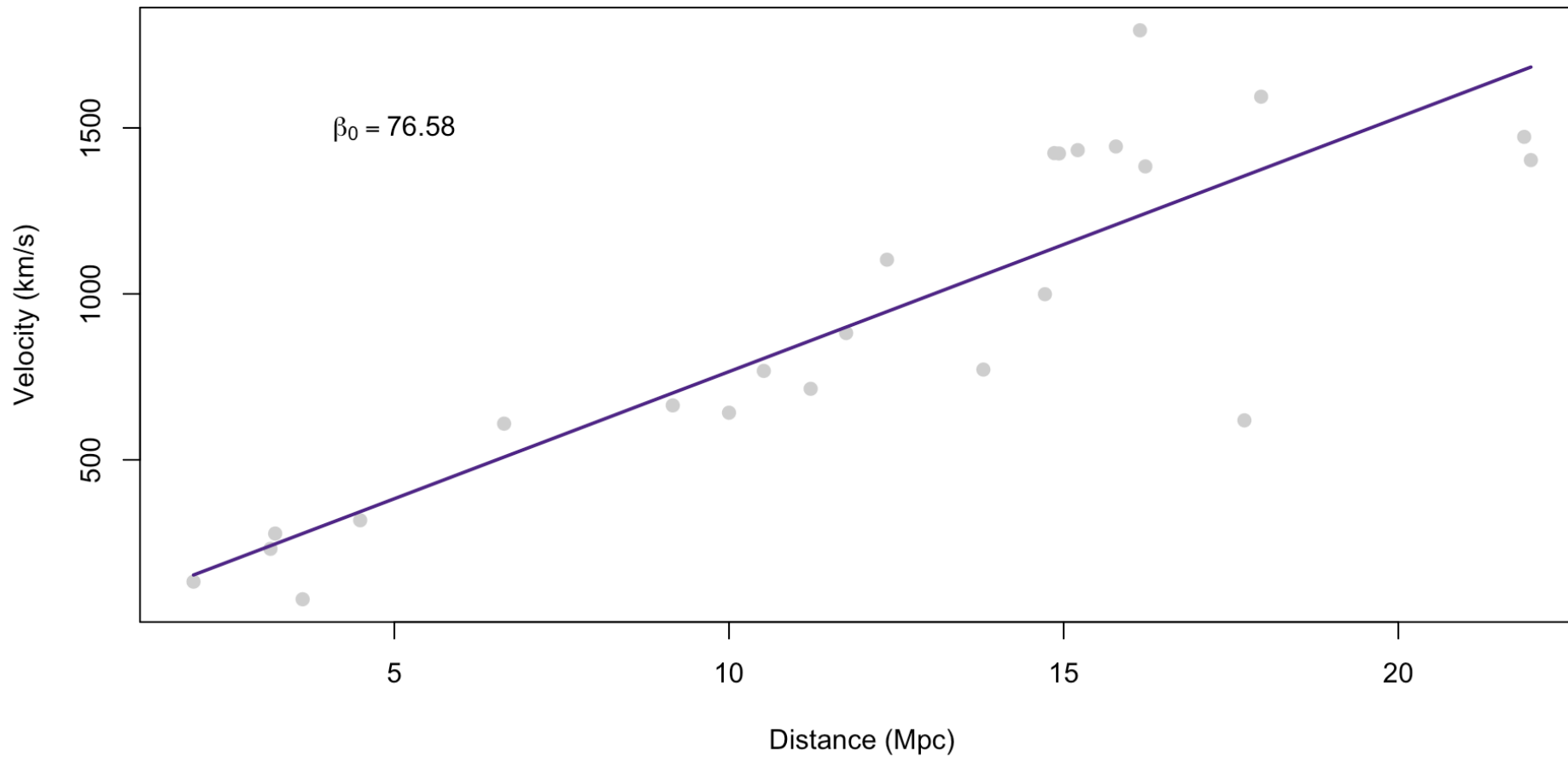


Point Estimates

- $\bar{x}, s^2, \hat{\beta}_0$
 - We've only calculated these so far
- They're **all** wrong estimates
 - How do we use a sample mean to infer the population mean?
 - Why did we spend 5 weeks on distributions and probability?

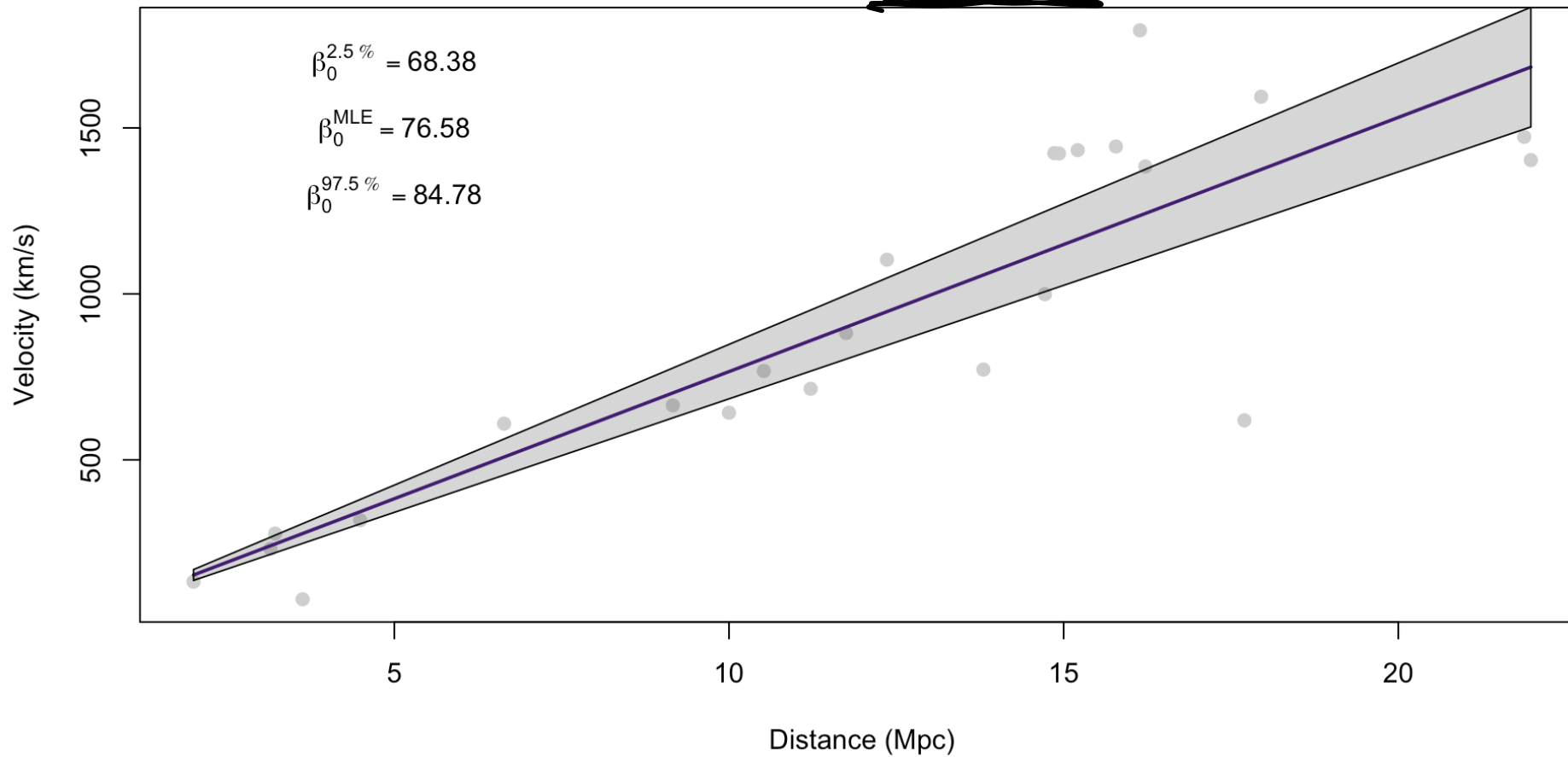


How old is the universe?

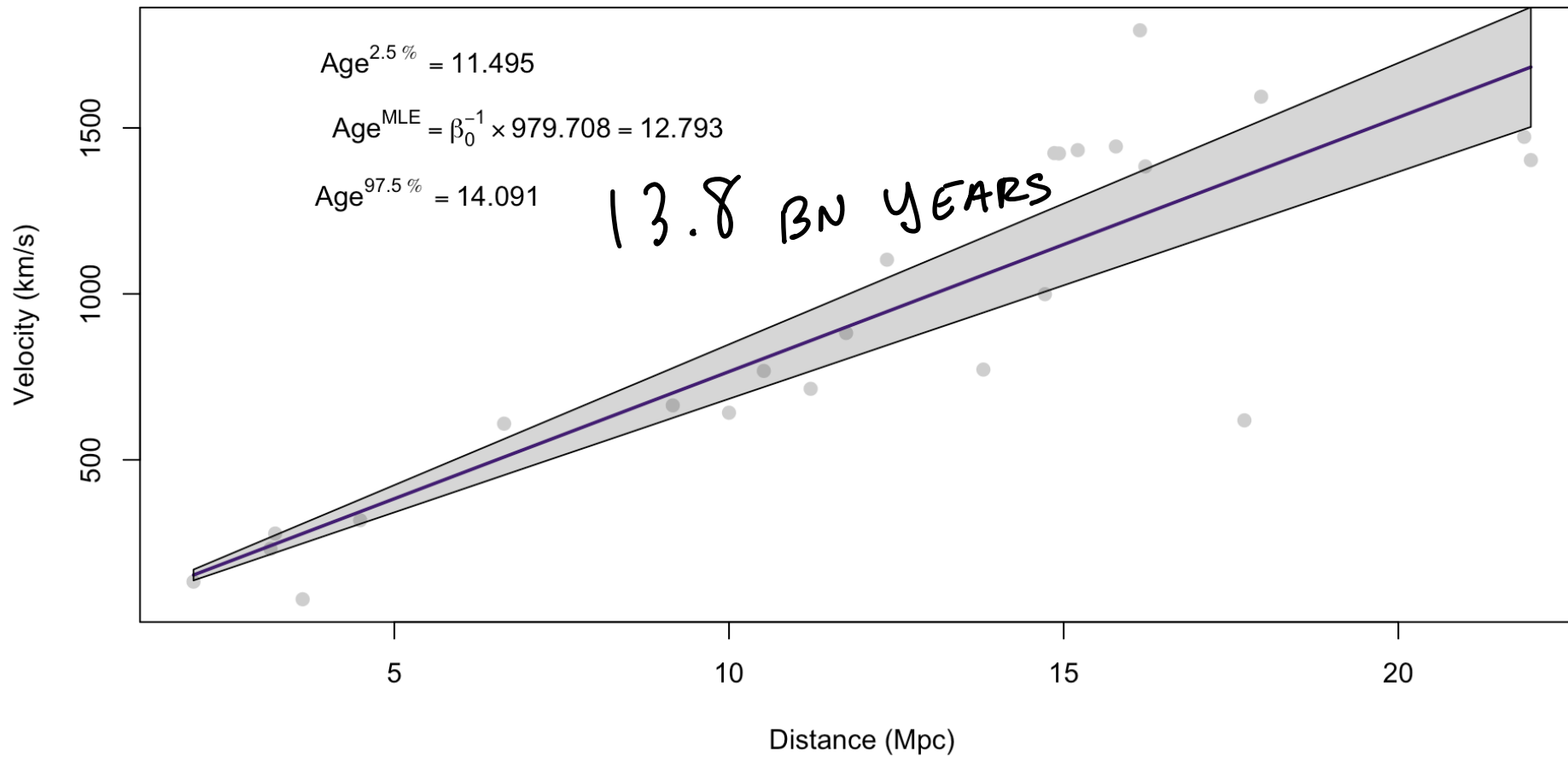


How old is the universe?

HETERO SCEDASTIC



How old is the universe?



Frequentist

Methods using the perspective that the probability of an event is defined as the relative frequency of that event's occurrence across *infinite* trials.

- Empirical probability
- “Data driven”
 - Data dependent
- Central Limit Theorem
- Simple in application



Bayesian

Methods using the perspective that the probability of an event is the result of comparing the likelihood and prior knowledge of the event with the observed evidence.

- Prior knowledge
- “Assumption driven”
 - Open to assumptions
- Horrendous in application



Linear Regression

- Frequentist:

$$y = \beta_0 + \beta_1 x + \epsilon$$
$$\epsilon \sim N(0, \sigma^2)$$

- Since $\mathbb{E}(\epsilon) = 0$:

$$y \sim N(\beta_0 + \beta_1 x, \sigma^2)$$



Linear Regression

- Bayesian model:

$$y \sim N(\beta_0 + \beta_1 x, \sigma^2)$$

$$\beta_0 \sim N(0, \sigma_{\beta_0}^2)$$

$$\beta_1 \sim N(0, \sigma_{\beta_1}^2)$$

$$\sigma^2 \sim \text{inv-Gamma}(q, r)$$



Bayes' Rule

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$



Interpreting Intervals

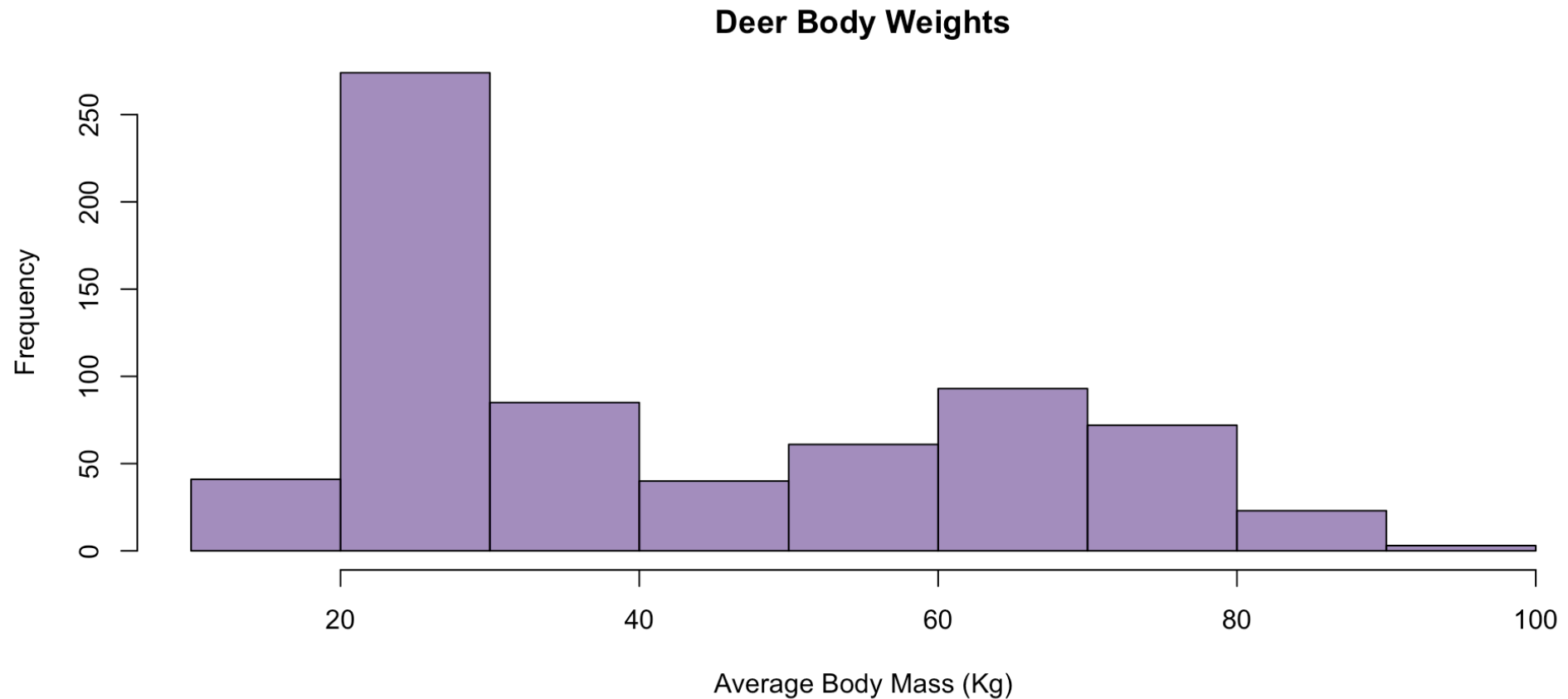
- Confidence Intervals
 - Frequentist
 - “the probability the method is going to produce the correct result.”
- Credible Intervals
 - Bayesian
 - “the probability that the estimate is the truth.”



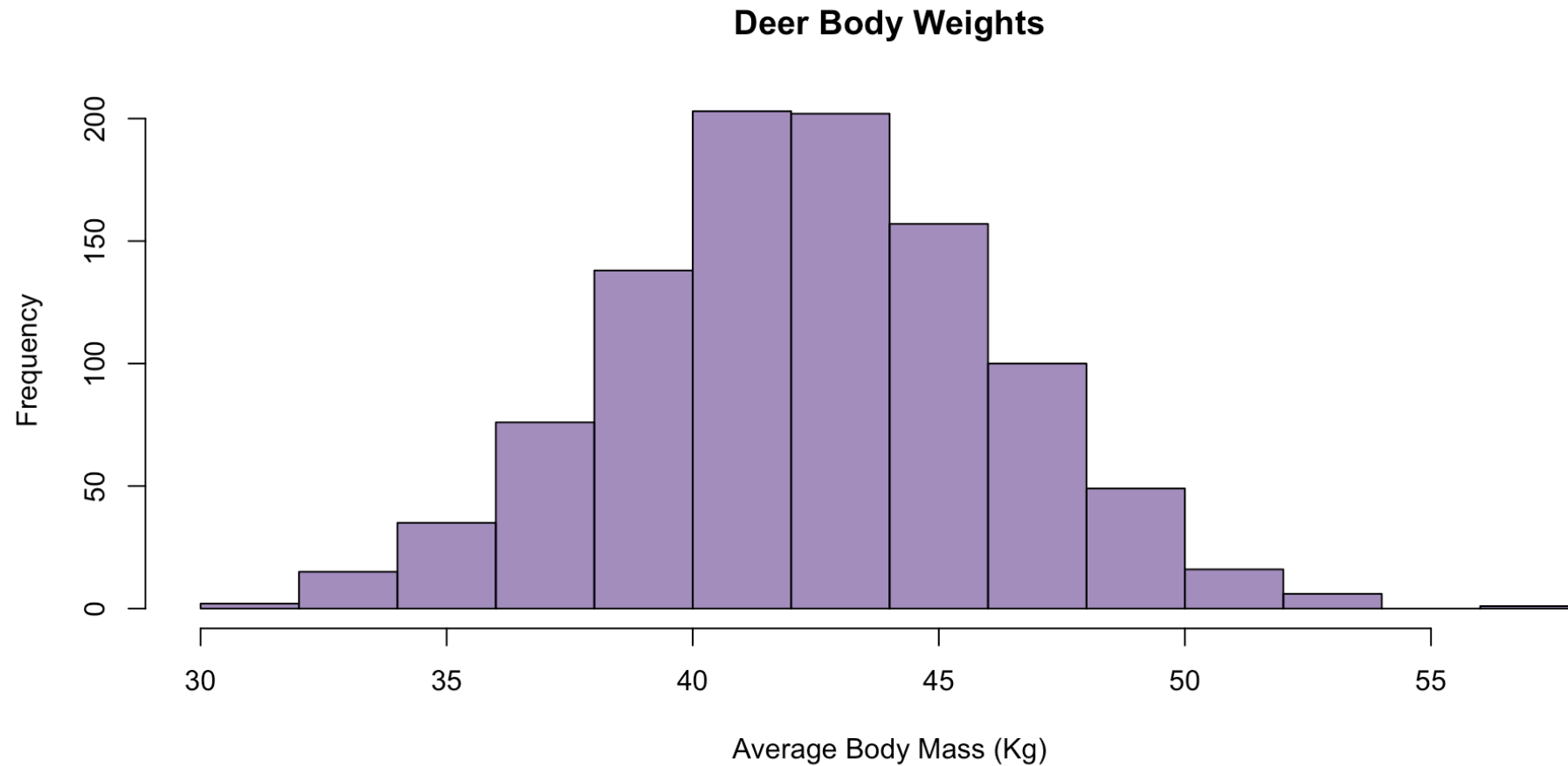
Applying Distribution Theory



Sampling Distributions



Sampling Distributions



Intervals

Given $\bar{x} = 53.17$, $s = 17.127$, and $n = 10$, calculate the 95% interval of deer body weights.

95% z-interval:

$$L = \mu - z\sigma$$

$$U = \mu + z\sigma$$

$$P(-z_0 < Z < z_0) = 0.95$$

$$\bar{X} \sim N(\mu_{\bar{x}}, \sigma_{\bar{x}}^2) = N\left(\mu, \frac{\sigma^2}{n}\right)$$



Intervals

$$\bar{x} \sim N(\mu, \frac{\sigma^2}{n})$$

- We need population parameters to proceed
 - How can we get rid of them? Fix the distribution to $N(0, 1)$?
- What could we do to fix a non-standard normal distribution?



Intervals

$$z = \frac{x - \mu}{\sigma}$$

- This still needs population parameters



Pivots

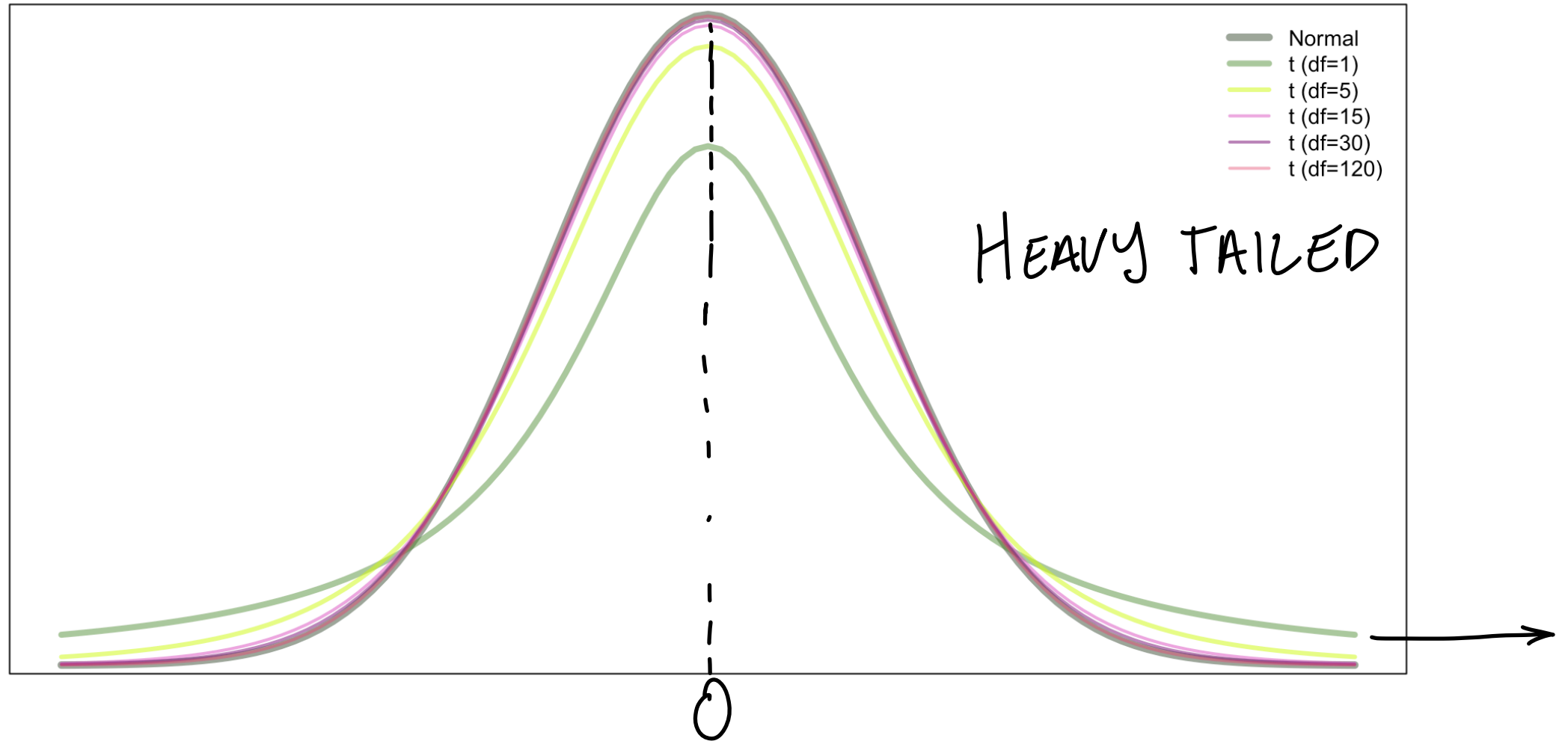
Pivotal Quantity (Pivot): Any function whose probability distribution does not rely on the population parameters of a given random variable.

$$t^* = \frac{x - \bar{x}}{s/\sqrt{n}}$$

- We can “free” the statistic from the population
- Then inference can come exclusively from samples



t-Distribution



t-Pivot

$$\bar{x} + t_0 \frac{s}{\sqrt{n}}$$

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$t(df)$$

$$P(-t_0 < T < t_0) = 0.95 \quad df = n - 1$$

$$P(-t_0 < T < t_0) = 0.95$$

$$P(-t_0 < \frac{\bar{x} - \mu}{s/\sqrt{n}} < t_0) = 0.95$$

$$P(-t_0 \frac{s}{\sqrt{n}} < \bar{x} - \mu < t_0 \frac{s}{\sqrt{n}}) = 0.95$$

$$P(\bar{x} - t_0 \frac{s}{\sqrt{n}} < \mu < \boxed{\bar{x} + t_0 \frac{s}{\sqrt{n}}}) = 0.95$$



Confidence Intervals

$$0.99 = 1 - \alpha$$

$$\alpha = 0.01$$

$$\alpha/2 = 0.005$$

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$P(-z_0 < Z < z_0) =$$

$$0.99$$

$$1 - 0.99 = 0.01/2$$



Confidence Intervals

Point estimate = $\bar{x}$

$$z_{\alpha/2} \sigma$$

$$df \rightarrow \infty \quad t = z$$

$$\text{Margin of error} = t_{\alpha/2} \frac{s}{\sqrt{n}}$$

Point estimate $\pm$ Margin of error

$\wedge$
 p



t-table

df	0.500	0.250	0.200	0.150	0.100	0.050	0.025
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228



Sample Mean Intervals

$$CI = \bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$CI = 53.17 \pm 2.262 \times \frac{17.127}{\sqrt{10}}$$

$$CI = 53.17 \pm 12.25107$$

$$CI = \begin{cases} 53.17 - 12.25107 = 40.91893 \\ 53.17 + 12.25107 = 65.42107 \end{cases}$$

$$CI = (40.92, 65.42)$$



Go Away

